**18** **A** When the two polarisers are parallel($\theta = 0$), the intensity of emergent beam is $I_0$
When Q is rotated at angle $\theta$, the intensity of emergent beam is reduced by 30%, which means final intensity should be $0.70 I_0$. Appy Malus Law,

$$0.70 I_0 = I_0 \cos^2 \theta$$

In the first quadrant, $\theta = 33°$

In the third quadrant, $\theta = 180° + 33° = 213°$

**19** **D** The electric field within the conductor will be zero(since no net charge/equipotential